

Systems for Design and Implementation : Assignment 2

BRONZE CHALLENGE IS NOT COMPLETE WITHOUT:

-unit tests and coverage for all the important CRUD operations

**-thorough client-side and server-side validation is performed
for any input**

SILVER CHALLENGE

-detect if the network is down or if the server is unreachable and enable offline support for the CRUD operations using local memory on the client device. When the connection is re-established, implement data synchronization with the server

-on the server side, create an endpoint that starts (and one that stops) an asynchronous loop which programmatically generates fake but valid entities using libraries such as [Getting Started | Faker](#) and adds them to the collection of entities maintained on the server side. Once a batch of items is created and added, the server uses websockets ([WebSocket and Its Difference from HTTP - GeeksforGeeks](#)) to alert the client which updates the master/detail view and the charts that are presented side by side according to the newly received information

GOLD CHALLENGE

BE AMBITIOUS! Reimplement the server side with graphql, i.e take the same logic and data that you already have, but now expose them through a graphql interface instead of classical endpoints

BE RESPONSIVE! implement infinite scroll behaviour on the front end based on the pagination on the back end. do it with the minimum usage of the network infrastructure (use prefetching of pages if necessary)

BE DILLIGENT! implement a 1-to-many relationship between objects from the domain of your application, fullstack, from the front-end to the back-end, covering all CRUD operations and basic statistics

Deadline for the bronze[5p] challenge: Week 5

Deadline for the silver[+2p] and gold[+3p] challenges without delay: Week 6[max grade 10]

Deadline for the silver[+1p] and gold[+1.5p] challenges with one delay: Week 7[max grade 7.5]

Deadline for the silver[+0.5p] and gold[+0.75p] challenges with two delays: Week 8[max grade 6.25]